

| COUNTRY |              |    |
|---------|--------------|----|
| int     | country_id   | PK |
| string  | country_name |    |
| string  | region       |    |
| string  | iso_code     |    |

| USER      |            |    |
|-----------|------------|----|
| int       | user_id    | PK |
| string    | name       |    |
| string    | email      |    |
| string    | role       |    |
| timestamp | created_at |    |

| DATASET   |              |    |
|-----------|--------------|----|
| int       | dataset_id   | PK |
| string    | dataset_name |    |
| string    | source       |    |
| int       | record_count |    |
| timestamp | created_at   |    |

| HDI_INPUT_DATA |                 |    |
|----------------|-----------------|----|
| int            | input_id        | PK |
| int            | user_id         | FK |
| int            | country_id      | FK |
| float          | life_expectancy |    |
| float          | education_index |    |
| float          | gni_per_capita  |    |
| timestamp      | submitted_at    |    |

| SESSION   |            |    |
|-----------|------------|----|
| int       | session_id | PK |
| int       | user_id    | FK |
| timestamp | login_at   |    |
| timestamp | logout_at  |    |
| string    | ip_address |    |

| ML_MODEL  |            |    |
|-----------|------------|----|
| int       | model_id   | PK |
| int       | dataset_id | FK |
| string    | model_name |    |
| string    | algorithm  |    |
| float     | accuracy   |    |
| timestamp | trained_at |    |

| HDI_PREDICTION |               |    |
|----------------|---------------|----|
| int            | prediction_id | PK |
| int            | input_id      | FK |
| int            | model_id      | FK |
| float          | hdi_score     |    |
| string         | hdi_category  |    |
| timestamp      | predicted_at  |    |

| VISUALIZATION_REPORT |               |    |
|----------------------|---------------|----|
| int                  | report_id     | PK |
| int                  | prediction_id | FK |
| string               | chart_type    |    |
| string               | file_path     |    |
| timestamp            | generated_at  |    |

